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Spectral Analysis Methodology

How we measure what a star is made of, starting with the Sun

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METHODOLOGY

Spectral Analysis Methodology

How we measure what a star is made of, step by step, starting with the Sun

SECTION 01

One star, A to Z

The question is simple to state. How much of each element is in a star's outer layers? Every element absorbs light at its own set of wavelengths. Those absorption lines show up as dark notches in the star's spectrum, the star's light spread out by wavelength. The depth and shape of each notch tell us how many atoms of that element are there.

We start with the Sun. Its composition has been measured carefully by other teams, so it is the test. If our method cannot reproduce the published solar values, it has no business measuring other stars. The same process will then be applied to stars that host planets.

THE SIXTEEN STEPS

Every star and every element goes through the same steps, in this order. Steps 1 to 8 are preparation. Nothing is measured until they are done.

1 **Choose the star.**

2 **Find every instrument and data holding.** A holding is one set of observations of the star from one instrument, as we store it.

3 **Audit each holding.** Check what the files really are: their units, their wavelength scale and whether anything has already been done to them.

4 **Correct telluric absorption and normalise the continuum, once per holding, then freeze.** The prepared spectrum is then used unchanged for every element.

5 **Collect the literature on the star.** What have other teams measured, and how?

6 **Secure the atomic line data.** Wavelengths and line strengths for every line we might use.

7 **Grade lines against the published solar sets.**

8 **Fix or solve the stellar parameters.** Temperature, surface gravity and the other numbers that describe the star's atmosphere.

9 **Measure iron first.** Iron has more usable lines than any other element, so it is the first check that everything above is working.

10 **Measure every band, instrument and engine separately.**

11 **Uncertainty analysis.** Build the error bar for each result.

12 **Check against the literature.**

13 **Report problem lines.** Lines that disagree with the rest are listed, not quietly dropped.

14 **Derived ratios.** Ratios such as carbon to oxygen, which matter for planets.

15 **Human review.**

16 **Publication.**

THE ORCHESTRATOR

The steps are run by one program, the orchestrator. We give it one command per star and element. It works through every combination of band, instrument and calculation method, and reports the status of each one: done, held back and why, or not run yet.

SECTION 02

Data

No single instrument covers every wavelength we need, so we use several. Each one is measured on its own. Having more than one look at the same lines also shows us when a result depends on the instrument rather than on the star.

WAVELENGTH BANDS

We split the spectrum into bands. Each band is handled under its own rules, because the problems change with wavelength. In the near-UV, lines are packed so tightly that no line stands alone. In the red and infrared, lines are sparse but Earth's atmosphere absorbs strongly.

BAND	WAVELENGTHS (Å)
Near-UV	3000 – 3780

BAND	WAVELENGTHS (Å)
VIS (visible)	3780 – 6910
Red-optical	6910 – 9199
NIR (near-infrared)	9199 – 13000
H (infrared)	15007 – 17494
K (infrared)	19452 – 24846

SOLAR HOLDINGS BY BAND

HOLDING	WHAT IT IS	BANDS USED
Kitt Peak solar atlas, our telluric correction	The 1984 Kitt Peak spectrum of sunlight, corrected for Earth's atmosphere by us with molecfit	Near-UV, VIS, red-optical, NIR
Kitt Peak solar atlas, Kurucz 2005 correction	The same atlas with Earth's atmosphere removed by Kurucz (2005)	Near-UV, VIS, red-optical, NIR
IAG solar atlas	A second, independent spectrum of sunlight from Göttingen, published already corrected (Reiners et al. 2016; Baker et al. 2020)	VIS, red-optical, NIR

HOLDING	WHAT IT IS	BANDS USED
HARPS	Ten direct exposures of the Sun from the HARPS spectrograph in Chile, corrected by us with molecfit	VIS
CRIRES+	Infrared spectra of sunlight reflected off the asteroid Vesta, from the CRIRES+ spectrograph in Chile	NIR, H, K

Raw spectra are never used. Every spectrum in this table has been corrected for Earth's atmosphere and normalised before any line is measured.

OTHER TARGETS

We also hold spectra of planet hosts and nearby stars, including 55 Cancri A and α Centauri A and B. These include HARPS, ESPRESSO, CRIRES+ and Hubble ultraviolet spectra. They wait until the Sun is finished. Each system page lists what we have for it.

SECTION 03

Telluric correction and continuum

Before a spectrum can be measured, two things have to be removed: the fingerprint of Earth's air, and the overall brightness level. We do this once for each holding. The result

is then frozen and used for every element, so all elements see the same spectrum.

TELLURIC LINES

Telluric lines are absorption by Earth's own atmosphere, mostly water vapour and oxygen. They land on top of the star's lines and can make a line look deeper than it is. Every spectrum taken from the ground is corrected. Only a telescope in space escapes them.

For HARPS and CRIRES+ we model the atmosphere on the night of each observation with molecfit, a tool from the European Southern Observatory (Smette et al. 2015). The Kitt Peak atlas has no record of when or where each part was observed, so we fit its atmosphere freely. We also use the independent Kurucz (2005) correction of the same atlas. The IAG atlas comes already corrected by its authors.

CONTINUUM

The continuum is the brightness the star would have with no absorption lines at all. Line depths are measured against it, so it has to be set to 1 first. This is called normalising.

Published solar studies place the continuum at the local high points of the spectrum near each line. We do the equivalent inside the fit. Near each line we compare the observed spectrum with our own calculated spectrum, and set the continuum level from that comparison. Points that disagree badly, for example because of an unlisted line, are clipped out first.

For weak lines this matters a great deal. A line only 1% deep is changed noticeably by a continuum error of a tenth of a percent. So we also repeat each measurement on the second, independent IAG atlas. The difference goes into the error bar as the continuum uncertainty.

In the near-UV the true continuum is never seen, because lines cover every wavelength. There we can only place a pseudo-continuum, and its uncertainty is priced separately.

SECTION 04

Atomic data and line grades

To turn a line's depth into a number of atoms, we need to know how strongly a single atom absorbs at that wavelength. This is the line's oscillator strength, written gf. A wrong gf gives a wrong abundance, however good the spectrum is. So we choose lines by how well their gf is known.

THREE LINE GRADES

GRADE	WHICH LINES	WHY
Reference	The lines used by the published solar analysis, with the gf values those authors used. That is Asplund, Amarsi & Grevesse (2021), or the paper they cite for that element. Any line with a laboratory-measured gf is added.	We repeat published work first, then extend it. If we measure the same lines with the same atomic data, any difference from the published answer comes from our spectra or our method, and we can find it.

GRADE	WHICH LINES	WHY
Codex	Any line whose gf comes with a published uncertainty: a laboratory measurement, a stated error, or a NIST accuracy grade of A to C. Lines up to 60% deep.	A wider set of lines, each with an error we can put a number on.
Deep	The same gf standard as Codex, for lines deeper than 60%.	Strong lines are more sensitive to the star's gas motions and to line broadening. They are reported, but as a secondary result.

A line whose gf has no stated uncertainty can still be measured, but it does not reach a grade. We note the gap rather than invent an error for it.

ONE GF PER PHYSICAL LINE

Many lines are really a cluster of very close components, split by the structure of the atom's nucleus (hyperfine structure) or by its isotopes. The calculation needs each component to get the line's shape right. But the gf is fixed once for the whole physical line, from a single master table. The components are scaled together so that their total matches it. A line with no entry in that table stops the run with an error. Nothing falls back to a default.

LINES WE REMOVE

Published solar studies leave out lines that are saturated, meaning so strong that the depth barely changes with abundance, and lines too blended with neighbours to measure. We do the same. A saturated line that the published solar analyses did not use is proposed for removal, and a person approves each one. Blends are handled by calculating every known atomic and molecular line in the window together.

SECTION 05

Model atmospheres and NLTE

To predict what a line should look like, we need a model of the star's outer layers: how temperature and pressure change with depth. This is a model atmosphere. We also need to know how the atoms in it behave. Different choices give slightly different answers, so we run several and report each one on its own.

MODEL ATMOSPHERES

A one-dimensional (1D) model treats the atmosphere as flat layers, each with one temperature. We use two standard 1D families: MARCS (Gustafsson et al. 2008) and ATLAS9 (Castelli & Kurucz 2003). The real Sun has rising hot gas and sinking cool gas side by side. A ⟨3D⟩ model is built from a full 3D simulation of that motion (the Stagger grid, Magic et al. 2013), averaged into layers. Swapping MARCS for ATLAS9 on the same lines is one of the measured terms in the error bar.

NLTE

The simplest calculation assumes LTE, local thermodynamic equilibrium: that each layer's atoms are set by its temperature alone. In a star's thin outer layers that is not quite true. Light from deeper down also excites the atoms. Allowing for this is called NLTE (non-LTE). It can shift an abundance by a few hundredths of a dex or more. (Abundances are on a log scale, and one dex is a factor of ten, so 0.01 dex is about 2%.) We include NLTE in two ways. The first takes published correction tables for each line (from Amarsi, Bergemann and collaborators) and applies them to the result. Where those tables stop, we compute corrections the same way from the published model atoms (Amarsi et al. 2020). The second puts NLTE inside the calculation itself. Turbospectrum (Gerber et al. 2023) reads departure coefficients, which say how far each atomic level is from LTE, and builds them into the predicted spectrum. That is done on both MARCS 1D and ⟨3D⟩ models.

TREATMENT	ATMOSPHERE	HOW NLTE ENTERS
1D LTE	ATLAS9 or MARCS	None
1D NLTE, correction tables	ATLAS9	Per-line correction from published tables
3D NLTE, correction tables	ATLAS9 base	Per-line 3D NLTE correction from published tables
1D NLTE, in the calculation	MARCS	Departure coefficients inside the synthesis

TREATMENT	ATMOSPHERE	HOW NLTE ENTERS
⟨3D⟩ LTE and ⟨3D⟩ NLTE	Stagger ⟨3D⟩	None, then departure coefficients, on the same atmosphere

Each treatment is its own result. They are never averaged together. If two treatments disagree, the disagreement is information about the physics, and averaging would hide it.

SECTION 06

Measuring an abundance

An abundance is the number of atoms of an element for every trillion hydrogen atoms, on a log scale. For the Sun, iron is about 7.46 on that scale (Asplund et al. 2021). We find it by changing the abundance in our calculation until the predicted line matches the observed one.

SPECTRUM SYNTHESIS, THE MAIN ROUTE

Spectrum synthesis means calculating the whole stretch of spectrum around a line: the line itself, its neighbours, and any molecules there. We use Turbospectrum (Plez 2012) through iSpec (Blanco-Cuaresma et al. 2014). The program changes the abundance until the calculated spectrum best fits the observed one. Because every neighbour is in the calculation, a blend is modelled rather than ignored. In the near-UV this is the only route allowed, since no line there stands alone.

PROFILE FIT, THE SECOND ROUTE

In the visible and red bands, many lines are well separated. There we can also fit the line's own shape and measure its equivalent width: the width of a fully dark strip with the same area as the line. A curve of growth, which links equivalent width to abundance, turns it into an abundance. The two routes are kept as separate results. Simply adding up the missing light over a fixed window is not used. Tested against the HARPS lines, it clipped the wings of strong lines and took in neighbours of weak ones.

FROM LINES TO ONE NUMBER

Each line gives its own abundance. A result combines the lines of one element, from one band, one instrument holding and one treatment. Its value is the median over those lines, so a single odd line cannot pull it far. When the published analysis we are repeating used a plain average, we use the same.

SECTION 07

Uncertainty

A number without an honest error bar cannot be compared with anything. Our error bar has two parts. The statistical part is how much the lines scatter. The systematic part is how much the answer moves when we change a choice we had to make.

STATISTICAL

With several lines, the statistical error is the standard error of the mean: the scatter between lines divided by the square root of their number. With one or two lines there is

no scatter to measure. Then we use how sharply the fit's quality worsens as the abundance moves away from the best value.

SYSTEMATIC

For each systematic term we re-run the same lines with one thing changed, and record how far the answer moves. Each result carries every term below. If any one is missing, the result is held back and no error bar is published.

TERM	WHAT WE CHANGE OR CHECK
Atomic data (gf)	The published gf uncertainties, combined as explained below
Temperature	The star's temperature
Surface gravity	The star's surface gravity
Metallicity	The star's overall metal content
Microturbulence	The small-scale gas motion that broadens strong lines, raised and lowered
Continuum	Two placements of the continuum, and the second, independent IAG atlas
Fit window	The width of the stretch of spectrum fitted
Pseudo-continuum	Near-UV only, where the true continuum is never seen

TERM	WHAT WE CHANGE OR CHECK
Telluric	What is left of Earth's atmosphere after correction, measured against an independent correction
Method check	How far the method misses the known answer in the visible band
NLTE	The uncertainty in the NLTE correction
Model atmosphere	MARCS swapped for ATLAS9
Hyperfine and isotopes	The splitting of a line into close components
Blends	Neighbouring lines missing from the calculation
Molecules	For a molecular line, its dependence on the other element in the molecule

Some terms do not apply to every result. Pseudo-continuum is zero outside the near-UV, for example. For the Sun, temperature, gravity and metal content are known so well that those terms come out at zero. Microturbulence is not, and is measured. The terms are then added in quadrature (as the square root of the sum of squares), together with the statistical part, allowing for any that move together.

GF ERRORS THAT MOVE TOGETHER

Laboratory gf values usually come from two measurements: how long an excited atom lives, and how its decay splits between lines. The lifetime sets an overall scale shared by every line from that source. If it is off, all those lines are off in the same direction, and

averaging more of them does not help. The other part of each line's error is its own, and does average down.

So for each gf source we record which kind of error it carries. The shared scale is counted once. The per-line part shrinks with the number of lines. A source we have not yet reviewed is treated the cautious way, as fully shared.

HOW ASPLUND ET AL. (2021) DID IT

For comparison, the published solar analysis used a weighted standard error of the mean, or the fit quality for one or two lines. To that they added three systematic terms in quadrature: half the difference between two model atmospheres, half the difference between full 3D and ⟨3D⟩, and half the NLTE correction (at least 0.03 dex where no NLTE calculation exists). They say atomic data errors were generally not included. Our budget is built the same way, with more terms, and includes gf errors.

SECTION 08

Publication and solar validation

A result is not published because it was computed. It has to pass a comparison with the literature and a human review first. Then it goes on the site, and anything it replaces is archived.

CHECKING AGAINST THE PUBLISHED SUN

For every element we collect the published solar values first, chiefly Asplund, Amarsi & Grevesse (2021) and Amarsi & Grevesse (2026), each with its method. Our result is compared with them. The visible band is the control. It has a known answer, iron at

7.46 (Asplund et al. 2021). A method that cannot reproduce it there has not earned the right to report a value in a band with no known answer. How far it misses is measured and goes into the error bar as the method check.

PROBLEM LINES AND HUMAN REVIEW

Lines that disagree with the rest, or fail a check, are reported for each run with the reason. Every proposed line removal, and every publication, is approved by a person. The program does the measuring. A person decides what is published.

ONLY CURRENT RESULTS COUNT

When a result is re-measured, for example after a correction to the atomic data, the new one replaces the old. The old one is moved to an archive with the record of why. Nothing is deleted, so any earlier number on this site can be traced.

SECTION 09

References

The full list, with links, is on the Sources page.

REFERENCE	PUBLICATION
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REFERENCE	PUBLICATION
Amarsi et al. (2019)	A&A 630, A104 — Carbon NLTE corrections
Amarsi et al. (2020)	A&A 642, A62 — NLTE departure coefficients for large surveys
Amarsi et al. (2021)	A&A 656, A113 — Solar C, N and O from molecular lines
Amarsi et al. (2022)	A&A 668, A68 — Iron NLTE corrections
Asplund, Amarsi & Grevesse (2021)	A&A 653, A141 — The chemical make-up of the Sun: a 2020 vision
Baker, Blake & Reiners (2020)	ApJS 247, 24 — The IAG solar flux atlas: telluric correction
Barklem & Aspelund-Johansson (2005)	A&A 435, 373 — Van der Waals damping constants
Bergemann et al. (2012)	MNRAS 427, 27 — Iron NLTE in 1D and ⟨3D⟩ model atmospheres
Bergemann et al. (2017)	ApJ 847, 15 — NLTE with 1D and ⟨3D⟩ models: magnesium
Blanco-Cuaresma et al. (2014)	A&A 569, A111 — iSpec spectral analysis framework
Castelli & Kurucz (2003)	Proc. IAU Symp. 210 — ATLAS9 model atmosphere grid

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Gerber et al. (2023)	A&A 669, A43 — NLTE radiative transfer with Turbospectrum
Gustafsson et al. (2008)	A&A 486, 951 — MARCS model atmosphere grid
Kramida et al. (2023)	NIST Atomic Spectra Database — physics.nist.gov/asd
Kurucz (2005)	High-resolution solar irradiance and telluric-corrected solar atlas
Lodders, Bergemann & Palme (2025)	Space Sci. Rev. 221 — Solar system abundances from the photosphere and CI chondrites
Magic et al. (2013)	A&A 560, A8 — The Stagger grid: averaged 3D model atmospheres
Pehlivan Rhodin et al. (2017)	A&A 598, A102 — Oscillator strengths of Mg I
Pehlivan Rhodin et al. (2024)	A&A 682, A184 — Transition data for Si I and Si II
Plez (2012)	Turbospectrum: Code for spectral synthesis — radiative transfer, native molecular lines
Reiners et al. (2016)	A&A 587, A65 — The IAG solar flux atlas
Ryabchikova et al. (2015)	PhyS 90, 054005 — VALD3 atomic line database

REFERENCE	PUBLICATION
Schmitt (2010)	Senior Thesis, University of Montana — predecessor 55 Cnc abundance study
Smette et al. (2015)	A&A 576, A77 — Molecfit: telluric absorption correction

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